The Ephemeris

Quarterly Edition April - June 2015
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President's Message

From Dave Ittner

First, I would like to thank Rob Jaworski for being an outstanding President these past two years. He guided this club through a number of difficult situations over his term and not only kept us on course but helped the club move forward. Thanks.

It is an honor to be selected to serve as the President of SJAA for this year. I am quite proud to be a member of SJAA, not just because we conduct so many events and programs for the public, or because we have so many member-only events. I am proud to be associated with SJAA because of its members. This club is what it is because of the fine individuals who freely give their time and effort to make this club so special.

When individuals band together to form a club, their combined individual contributions make a huge impact. All we have to do is identify what we want to accomplish and then set out to achieve it. Take for example the need we have for Internet at Houge Park. Sukhada P. stepped up and took on this task and made it happen. Yes – we now have high speed internet for our members at Houge Park! Huge thanks go to Sukhada for that!

My goals for this year are to assist with improving the operations of the club, surveying current and past members, recruiting volunteers, and finding ways to make the club more social. I'm always open to input and feedback, whether they are new ideas or your opinions on how things are going. Please do let me know your thoughts.

And most importantly of all, I look forward to getting to know as many of you as possible. Please come out and join us, either at the meetings, or in the hills, preferably both.

Yosemite star party, August 21-22, 2015

From Jim Van Nuland

The annual SJAA Yosemite star party will be held on August 21 & 22, at Glacier Point in Yosemite National Park. Up to 30 people will be given free admission and camping, in exchange for two public events on Friday and Saturday evenings. In what time is left, we can be tourists. We are expected to have at least one scope per two people, and to attend both star parties, not just Friday or Saturday.

The camping is rough by modern standards: no dining room, no showers, no hot water, no sidewalks. There are a few motels in the area; google on "Yosemite West". It's an unincorporated area about 17 mountain miles from Glacier Point. Don't be surprised if all are fully booked. For all these reasons, this is probably not suitable for a family camping trip.

Note that this is not an SJAA club star party; rather, it is a Park star party. We are "paid" by free entry and campsites. If you do not have experience with setting up for the public, this is not the place to start. We will have 200 to 300 customers, most of whom have never looked through an astronomical telescope before.

Read about it on the SJAA Yosemite page <http://www.sjaa.net/school/yosemite.htm; also read the Yosemite FAQ page at <http://www.sjaa.net/school/yosefaq.htm>. Then contact me with remaining questions. If you can tolerate the limitations, e-mail me at <jvn@sjpc.org>. Tell me the number of people you'll have, and the number of scopes that will be set up for the public. Send along a mailing address; I'll mail gate passes about 3 weeks before. Priority is given to SJAA members; after about 6 weeks, I will consider non-members who have experience with public presentations.

June - Aug 2015 Events

Board & General Meetings
Saturday: 6/27, 7/25, 8/29, 9/26
Board Meetings: 6 -7:30pm
General Meetings: 7:30-9:30pm

Solar Observing (locations differ)
6/13, 7/5, 8/2, 9/13

Fix-It Day (2-4pm)
Sunday: 6/7, 7/5, 8/2, 9/6

Houge Park In-Town Star Party
Friday: 6/12, 6/26, 7/10, 7/24, 8/21, 9/4

Intro to the Night Sky Class (Houge)
Friday: 6/26, 7/24, 8/21, 9/18

RCDO Starry Nights Star Party
Saturday 6/6, 7/18, 8/8, 9/5

Binocular Star Gazing
Saturday 7/25, 8/22

Imaging SIG Mtg
Tuesday, 6/16, 8/18, 9/15

Quick STARt (by appointment)
Saturday: 6/13, 8/8

Unless noted above, please refer to the SJAA Web page for specific event times & locations.

SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.

INSIDE THIS ISSUE

Science

The Hubble Space Telescope Anniversary - 25 YEARS

From the dawn of humankind to a mere 400 years ago, all that we knew about our universe came through observations with the naked eye. Then Galileo turned his telescope toward the heavens in 1610. The world was in for an awakening.

The Hubble Space Telescope is named in honor of astronomer Edwin Hubble.

That nebulous patch across the center of the sky called the Milky Way was not a cloud but a collection of countless stars. Within but a few years, our notion of the natural world would be forever changed. A scientific and societal revolution quickly ensued.

In the centuries that followed, telescopes grew in size and complexity and, of course, power. They were placed far from city lights and as far above the haze of the atmosphere as possible. Edwin Hubble, for whom the Hubble Telescope is named, used the largest telescope of his day in the 1920s at the Mt. Wilson Observatory near Pasadena, Calif., to discover galaxies beyond our own.

Hubble, the observatory, is the first major optical telescope to be placed in space, the ultimate mountaintop. Above the distortion of the atmosphere, far far above rain clouds and light pollution, Hubble has an unobstructed view of the universe. Scientists have used Hubble to observe the most distant stars and galaxies as well as the planets in our solar system.

Hubble's launch and deployment in April 1990 marked the most significant advance in astronomy since Galileo's telescope. Our view of the universe and our place within it has never been the same. Since then, Hubble has reinvigorated and reshaped our perception of the cosmos and uncovered a universe where almost anything seems possible within the laws of physics. Hubble has revealed properties of space and time that for most of human history were only probed in the imaginations of scientists and philosophers alike. Today, Hubble continues to provide views of cosmic wonders never before seen and is at the forefront of many new discoveries.

Here are a few of Hubble’s discoveries:

The Hubble Deep Fields: Hubble has observed the furthest away galaxies and the most ancient starlight ever seen by humankind

Age and size of the Universe: Hubble has calculated the age of the cosmos and discovered the Universe is expanding at an ever faster rate

The lives of stars: Hubble has revolutionized our understanding of the birth and death of stars.

The solar neighborhood: Hubble has taught us about planets, asteroids and comets in our own Solar System.

Exoplanets and proto-planetary discs: Hubble has made the first ever image of an exoplanet in visible light, and spotted planetary systems as they form.

Black Holes, Quasars, and Active Galaxies: Hubble found black holes at the heart of all large galaxies

Formation of stars: Hubble observes stars as they form from huge dust clouds

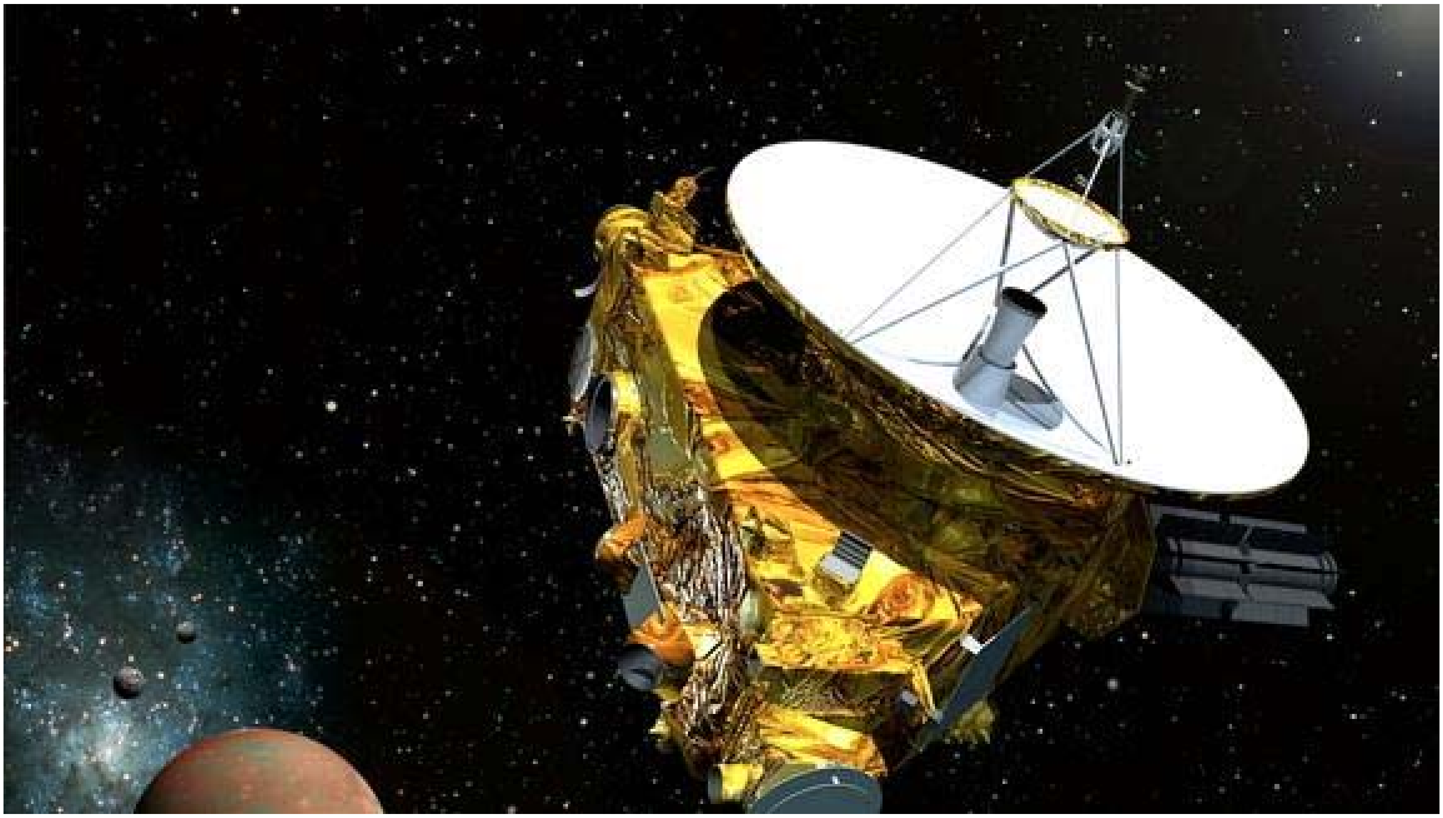
Composition of the Universe: Hubble studied what the Universe is made of, and came to some startling conclusions

Gravitational lenses: Astronomers use a helping hand from Einstein to increase Hubble’s range.

Credit: NASA.gov

Science

NASA Pluto Probe Begins Search for New Moons, Ring



Artist's concept of NASA's New Horizons probe zooming through the Pluto system in July 2015.

A NASA spacecraft speeding toward Pluto is casting a wary eye on the dwarf planet system, looking for anything that could trip it up in the home stretch of its historic mission.

NASA's New Horizons probe, which is set to perform the first-ever flyby of Pluto on July 14, has begun hunting for possible rings and undiscovered moons, in an effort to identify potential hazards near the dwarf planet.

Pluto is known to have five moons: Charon, Nix, Hydra, Kerberos and Styx. Charon, the innermost moon, was first spotted in 1978; at 648 miles (1,043 kilometers) in diameter, it's nearly half as wide as Pluto itself. The other four satellites are all tiny, and were discovered by NASA's Hubble Space Telescope between 2005 and 2012.

Spencer said he wouldn't be surprised if New Horizons did indeed add some more names to this list.

Credit: space.com and NASA.gov

SJAA Member Observation Posts

Editor's Note: The following 6 pages of Club member Observing reports / comments were all pulled from the SJAA Group Yahoo page.

I Saw Saturn!
May 5, 2015

From Bic Tran

I know for old hands in the club, seeing Saturn is Meh. But for a newbie like me, the sight of Saturn last night was awesome; it was so clear with the thick ring around. Unaided, Saturn looked like a dim yellow star below on the left side of the moon But in my 8" Dob, it was truly glorious more so than Jupiter. I only saw a small point light in the upper corner, not sure it would be a Saturn moon. Would someone clarify on the Saturn moons?

From Jim Van Nuland

> Bic Tran wrote:
> Hi all,
> I know for old hands in the club, seeing Saturn is Meh..

Meh? /Meh???/ No, no, no, a thousand times, no!
I first saw Saturn in a scope in about 1953. I'm anxiously waiting for it to re-enter the evening sky.
If Saturn was the only thing I could observe, I'd still want to own a telescope. (I'd need only 4 eyepieces instead of 10....)
The Hubble photos are utterly awesome, but (see previous paragraph).
At school star parties, we all show different objects. But as the kids leave, I look around, and all the scopes are swinging onto Saturn.
Clear Skies!

From Rob Jaworski

Other than the moons, the other cool thing to note about Saturn is the tilt of its rings. In 2009, we were looking at Saturn pretty much edge-on, so that the rings nearly disappeared. In 2017, they will be at maximum tilt and Saturn will be showing off its northern hemisphere, so between now and then, the ring show will get even better.
And yes, if the conditions are very good, Saturn is simply stunning, you can almost get a 3D effect. I can stare at it for hours.

Death Valley National Park
From Sukhada Palav
April 10, 2015

Well this isn't technically an OR but I just got back from a wonderful trip from Death Valley National Park and thought I would share a little bit of observation I did from there.
It was a full moon weekend so the park had a ranger led full moon hike scheduled on Saturday, April 4th. I was expecting just a walk in moonlight on sand dunes but it turned out to be lot cooler than that! We met at the trailhead for Golden Canyon and took an hour long walk in narrow walls of the canyon with ranger Laura. There were about 50 people, lots of families with kids. Even if ranger distributed red paper to put on our flashlights, we had plenty of white light in addition to the moonlight. :-) It was a bit annoying at first but given most people there had never been out in the dark, their need to use the flashlight unnecessarily was probably understandable.

Golden Canyon itself is a really cool place. The rocks there have same formation as the rocks on the moon! Actually, there is this specific kind of rock found on the moon and it had a complicated name to remember (;-)) but Golden Canyon is the only place on the earth where that rock is found. Laura gave us a lot of information about similarities between this canyon's geology and moon's formation. It was almost like walking on the moon, as she put it, plus six times more gravity. She distributed some maps of moon. Then she demonstrated how moon's phases change and eclipses happen, posing kids in different positions. It was quite engaging. Kudos to her for managing that crowd all by herself in the dark! Golden Canyon also happens to be the place where a lot of NASA research happens and one of the Star Wars movies, Return of the Jedi was shot right where we were standing! As we progressed in the canyon, moon rose higher in the sky...lighting the walls of canyon in shimmering silver first and eventually, us. It was a beautiful night filled with a wonderful experience! I will highly recommend this full moon ranger led program to anyone who visits Death Valley national park.

On night of April 5th, we had a tent cabin accommodation in Panamint Valley. I was really excited about it because it is as close to camping as it could get in this trip. So after dinner, I took my binoculars and stepped out for a brief observation with my friends. Moon was shining fully bright so it wasn't an ideal situation. The lights from motel and restaurant across the street were not making it any better. But we took a few minutes' walk to get as a way as possible and managed to view some beautiful objects including M42, Jupiter, Beehive cluster, M45 and the moon. It made me wish I was there on a new moon night. I can't possibly imagine how beautiful the sky must look from there if you took the moon away. We couldn't see that legendary view of Milky Way from there. :-(Well, there is always the next time.... And hopefully next time, I will drive down and not fly so that I can also carry my telescope with me.

SJAA Member Observation Posts

Recreating the Herschel's Legendary "Night of Discovery"

The Hershel Sprint From Mark McCarthy
March 18, 2015

Prologue:

This month marks the end of the first year of my stargazing adventure. Almost everything I look at is new, and a thrilling discovery to me. So when I saw an internet forum post referencing the *S&T* article on William Herschel's "Night of Discovery" of April 11, 1785, during which he discovered 73 galaxies, I thought to celebrate by replicating his observation session, known as the "Herschel Sprint." As an extra challenge, I decided to approximate Herschel's experience as closely as I could.

William Herschel discovered barred spiral galaxy NGC 4911 on April 11, 1785. NASA / ESA / Hubble Heritage Team (STScI / AURA)

S&T posted a list of all the objects to be observed on their website. They suggested "purists" could make the sweep more authentic by using the drift method — fixing the telescope at meridian and only moving it 2-3° in altitude, letting the objects drift by as Herschel was forced to because of his mount. One could approximate his view by using an older style eyepiece with 45-50° apparent field of view (AFOV) at 150x. But why stop there, I thought? His telescope used an 18.7-inch speculum mirror, which (if Wikipedia is to be believed) has 66% of the reflectivity of a modern mirror, equating to an approximate effective aperture of 12.5 inches. I happen to have a 12.5-inch f/7 scope! I could use the 15-mm RKE eyepiece from my Astroscan to match the 45° AFOV, at 148x in the 12.5-inch. Best of all, I had previously observed only five of the objects on the list. These five objects (NGC 3245, 3486, 3504, 3277, and 3414) were observed on February 16, 2015 using a 20-inch telescope. Unless I looked in my notes, I had no memory of what they looked like; besides, they would appear significantly dimmer in my 12.5-inch scope. So, 68 of 73 objects would be completely new to me, and 5 of 73 would be "practically" new.

Incidentally, I'm now one month younger than Herschel was in 1785, so we have our ages in common too! I remember reading in a 19th-century biography that Herschel intensified his deep-sky observing when he found himself "on the wrong side of 45," as Caroline Herschel is said to have put it. This more or less explains the drive behind my observing.

I took care to preserve my ignorance, such as it was, by not doing any new research about Herschel's night or the objects to be observed. I downloaded the spreadsheet list of objects from the *S&T* website, but deleted all of the columns except the NGC number, RA, and Dec. — other columns contained some hints, like "low surface brightness," so I avoided them.

I wanted some way of confirming the object I was looking at during the sweep. At first I thought of navigating by star-hopping in the eyepiece; so I photocopied the relevant pages from *Uranometria 2000.0* and traced lines between each object on the list to plot my course. But I soon realized I could easily get lost; estimating how many degrees to move in the eyepiece over a several-hours-long session would be confusing. Then I remembered from my prior reading that Caroline Herschel, as she took dictation, had next to her an accurate clock set to sidereal time, and the telescope had a contraption using ropes, pulleys, and a barrel top marked in altitude degrees. As she took dictation, she could note an object's right ascension from the clock and the declination from the scale on the barrel. I didn't think it would be cheating to use modern equivalents: a digital clock and a digital inclinometer. I could find a star at 0° declination that was also preceding the first object on the list. When this star reached meridian, and with it centered in the eyepiece, I could set the time on the clock to the star's sidereal time and set the inclinometer mounted to my tube to zero. Thus calibrated, I could move my scope in altitude only to the first list object's declination and wait for it to pass through. So long as I didn't move in azimuth, these guides would remain accurate enough throughout the

Continued on Page 6

SJAA Member Observation Posts

session and help me find the objects.

The *S&T* spreadsheet showed right ascension in tenths of minutes, and declination in seconds of arc; I converted these into minutes & seconds for right ascension, and tenths of degrees for declination, to match what I would see on the digital clock and the inclinometer, and printed it out to use at the telescope. I kept my *Uranometria* charts as back-up, but did not refer to them to find the list objects.

This would be a severe test of my visual descriptive skills, which — since the objects pass through the FOV in less than a minute — would require great economy of perception and expression. There wouldn't be enough time to dwell on an object, search for fine detail, or write and sketch my observation as I normally do, but I did have a digital voice recorder that I velcroed near the focuser for dictation.

What would run through my mind as the pageant unfurled before me? All that was left was to wait for a cloudless, moonless night.

Report:

I arrived at Fremont Peak Observatory Association's site at Fremont Peak State Park, California, at sundown and was observing soon after. I used the 15-mm RKE from the beginning, because I wanted to become accustomed to its FOV and how objects would appear. I ran through some H400 objects, and could tell right away the night ahead would be more challenging than I thought. My February session at the Peak under similar conditions was with my 20-inch, and I recall all my observations spoke of finding objects with direct vision, with averted vision helping to reveal some fine detail. This time, objects were still bright enough to be found but averted vision (AV) was more necessary to bring out more of the object, such as a galaxy's halo or mottling. Only a handful of the Sprint objects are plotted in my *Pocket Sky Atlas* — which is targeted for small and medium size scopes and includes all the Messier, Herschel 400, and Caldwell objects. I was still able to find the H400 objects (mostly in Leo's hindquarters) perfectly well and enjoy them, but the comparative dimness with the 20-inch was a foreshadowing of how the Sprint session would go.

I used *Stellarium* beforehand to find a preceding, 0° calibration star that was bright enough for me to find. I used HIP48413 (RA 9^h 52^m 59^s, Dec 0° 00' 13.6"), which according to *Stellarium* would reach meridian at 11:14:35 p.m. PDT on 3/18/15. It was an easy find off Iota (ι) Hydrae. As it neared meridian I kept it in the center of view and monitored my watch; at the right moment I set the inclinometer to zero, and set my clock (an alarm clock with red numerals) to 9:53. Then I moved the scope up to 27.7° to wait for the first object, **NGC 3196**. But while I was waiting the inclinometer's backlight went off. I thought the unit itself turned off; when I pressed the "on" button, the unit reset zero to default bubble level zero! I went back to my calibration star and reset to zero again, but since I lost the sidereal time I skipped NGC 3196 and went to NGC 3245, which I found by star-hopping with the chart. Fortunately the declination angle matched what it should be. I held the galaxy in view until the sidereal time caught up with **NGC 3245**; I then stopped my azimuth movement and started the Sprint. NGC 3245 was one of the objects I observed in February with the 20-inch. My description then was: "205x long bright nucleus running N-S, small diffuse halo, 2.3' x 1' visible. Easily found and bright." Now with the 12.5-inch it was: "148x elliptical, glowing nucleus with DV, bright core with AV; 3:1 N-S. In a triangle of stars. Somewhat difficult to find." I saw less of the galaxy, and it was dimmer. The Sky Quality Meter w/Lens (SQM-L) reading in February was 20.71; now it was 20.60 or so — about the same. This was to be one of the brightest objects I was to observe during the Sprint.

I abandoned the thought of doing a pure sweep. Because I had to navigate, and because I wanted some way of following the list, I decided to simply move the scope to the altitude of the next object on the list, and monitor the clock to wait for it to pass into view. My clock didn't display seconds so it was difficult to judge time precisely. Often the objects would be at nearly the same RA time but at different declinations, and I would spend too much time on one and miss the next. But the main problem was the difficulty in seeing the objects to begin with — I didn't know how the object would appear, so I was scanning and using AV and just trying to see if there was anything in view. I sometimes suspected my calibration was slightly off, and I was like a ship whose bearing was wrong by a slight angle, causing it sail further off course over time. But from time to time I would find an object at the listed location, and it would restore my confidence in my setup.

Certainly Herschel's sky was not affected by light pollution, so he may have been able to see more at comparable apertures — we use larger apertures and higher altitudes to compensate for our light pollution. But I think my difficulty lay with my eye, which hasn't been fully trained to see dim objects. I haven't been at this long enough.

In the end I counted 28 "did not find" objects. 15 I regard as tentative – where my verbal description was not quite confirmed by my later check of the NGC/IC Project or Aladdin websites, which I use post-session to confirm my observations. 30 objects I can confidently confirm. My descriptions are composed primarily of the adjectives "very faint," "exceedingly dim," and "difficult." I don't want to leave the wrong impression: I was really enjoying myself. Any night out under the stars is a good night, and here I was stretching my abilities and having fun. There were some very good moments. My most exciting sight was **NGC 4169**, **4173**, **4174**, and **4175**, a U-shaped cluster of galaxies known as The Box. It was incredible to see them drift across the view, one galaxy revealing itself after the next. I will admit, though, my biggest disappointment was the **Coma Cluster**. I had anticipated I would see a flurry

Continued on Page 7

SJAA Member Observation Posts

William and Caroline Herschel discovered some 20 new galaxies in the heart of the Coma Cluster (Abell 1656).
POSS-II / Caltech / Palomar Observatory

of galaxies fly past my little window on the universe, and I was really looking forward to it. As it turned out, I missed the whole thing because I had missed one of the leading galaxies, which put my timing off for the rest. Since in this game there was no going back, I had to let them go.

The total Sprint takes about five and a half hours. There are gaps of 10, 15, or 20 minutes here and there where Herschel did not find any new object. He most likely kept working, but I took breaks. Finally, after nearly three and a half hours, just after yet another exceedingly dim glow in **NGC 5263**, the great globular cluster **M3** (NGC 5272) drifted into the eyepiece from the east in all its glory. I checked my inclinometer and my clock, and was very surprised to find I was only 0.2° off in my declination and ~30-60 seconds off in RA — little enough that M3 still floated cleanly into the FOV. It proved to me that my navigation was accurate, and the problem was the challenge of the objects and my own abilities.

M3 is not one of Herschel's discoveries, but I am sure he would have let it pass through his eyepiece, and enjoy, as I did, feasting on a big, bright object after such labor. Full circle: M3 was the first globular I ever saw, from Monte Bello Open Space Preserve at the end of last April, found with the help of a kind fellow observer.

There is a long gap from M3 to the next object on the list. On the *Uranometria* charts there are a few NGC objects, but I see from the NGC/IC Project website they were discovered by others. There were also several Abell galaxy clusters, too dim for even Herschel to find. It was past 3am and I was tired, so I skipped forward and star-hopped to the last two galaxies. I trained my scope on the **Corona Cluster,** just to see how it would look at this aperture and decide whether Herschel had a chance to see it. I detected some grayish mottling, but I saw no distinct object; this is probably what Herschel saw too, and could not recognize it for what it was. But all those dim ovals he could see, not clouds but with some structure, must have given him an inkling these were not simply "nebulae," these were a different class of object altogether. If only he knew what we know now!

After so much dimness I craved a big bright galaxy, so I viewed **M51**. I could see both galaxies, and the extension between them, and with AV could trace spiral arms coiling in and out of the bright center. That was very satisfying.

I ended the night with a bang: **Nova Sagittarii 2015 No. 2**, which was just coming out over the hills at ~4:30am, found with the aid of the AAVSO chart. I was pretty worn out and didn't try a magnitude comparison, but it was the brightest star in my binocular field. I later found its current estimate is magnitude 5.2.

I want to try the Sprint again, but next time with the 20-inch. I'll find a way to restrain that scope's altitude movement within the 3° band, so that I can leave the list behind and sweep as Herschel did, and enjoy the view to the fullest. On the bright side, with so many objects left unobserved on the list, it will still be a night of discovery for me.

- See more at: http://www.skyandtelescope.com/observing/observing-report-the-herschel-sprint04012015/#sthash.tW76VIXH.dpuf

Editor's Note: Mark's above report was posted on skyandtelescope.com, April 1, 2015. Way to go Mark!

Revisiting the Herschel Sprint
From Mark McCarthy

Posted on skyandtelescope.com April 22, 2015

The night of April 16, 2015, I made my second attempt at a drift method Herschel Sprint. I made a few changes from my March 2015 attempt. I used a 20-inch f/5.25 telescope, so I would have every chance to see all the objects on the list. I used a zoom eyepiece which when set at 18mm yielded 148× and 0.3° true field of view (TFOV). I used HIP48413 (RA 9h 52m 59s, Dec 0° 00′ 13.6″) as my calibration star, as before.

(Continued on Page 8)

SJAA Member Observation Posts

The time of its meridian transit, according to Stellarium, was 9:20:30 p.m. PDT. But, rather than using a clock and a list to follow each object's right ascension, I pressed the "record" button on my digital recorder the moment the calibration star was at meridian and centered in the eyepiece. I hoped this would allow me to simply record my observations without removing my eye from the eyepiece; so long as I said "centered" when the object being described was centered in the eyepiece, the recording would give me a fairly accurate elapsed time which I could later compare to the object list and charts to confirm what I saw. Using my inclinometer to find the upper and lower limits of the sweep's declination band, I used C-clamps to attach wood blocks to the altitude bearing on my telescope to restrict altitude movement within the band. I was thus freed from worrying about navigation and could concentrate on observing and describing, as Herschel could. All I had to do was gently, so as not to disturb the telescope's collimation, bob the telescope up and down in an even tempo to cover all of the sky within the band.

I once again observed from the Fremont Peak Observatory Association's site at Fremont Peak State Park (elevation 2700'), California. The temperature dropped into the 40s after dark and it was windy. Seeing was 5/7 and transparency 4/5; SQML was 20.60.

It's very difficult for me to untangle my observations and report my results. I spent a good part of this weekend transcribing my recording and noting the elapsed time of each of my comments. When I review these times versus what the elapsed time of the Sprint objects would be from my calibration star, I have very few matches. **NGC 3274** is a certain match: according to the Sprint list it would appear at 00:39:18 elapsed time. My comment at 00:39:14 on the recording is "close to center, there's a triangle of stars, between two at the base there's cloudiness, a galaxy, E-W elongation" which closely matches what I find on the NGC/IC Project photograph and description. But few of the others have an obvious corresponding match.

One reason was the unevenness of my altitude movement tempo. Before starting I timed my calibration star's transit across the eyepiece from east to west; it took 74 seconds. To cover the band completely I tried to move the telescope up in the band in 20 seconds, then down in 20 seconds, and so on. This should cover the band in altitude in thirds of eyepiece FOV. But the sky moves quickly through the eyepiece at that pace, leaving very little time to perceive and describe what one sees. Very often I could not keep that pace, or I might linger on an object just to make an observation of it, spoiling my timing and making me miss small swaths of sky, which incrementally added up to a lot of missed sky. Often I would see a galaxy just at the western edge of the FOV, having nearly missed it, and have just a few seconds to describe it. The swiftness of an object's passage through the FOV is evidenced in the poor quality of my descriptions, which for the most part are brief and just describe the overall impression, and are not helpful to confirm the object from a photograph or another's description later. As a result, most of my observations are unidentified.

Since my first attempt in March, I read *Discoverers of the Universe* by Michael A. Hoskin. In it I learned that William's brother Alexander had fitted the telescope with a mechanism which would ring a bell when the telescope was at the top and bottom limits of a sweep's declination band, to give William's assistant, pulling on the ropes, cues when to raise or lower the telescope. I wondered whether William had the assistant use a metronome to help maintain an even tempo — William was a musician, after all. However, metronomes did not come into use until the early 19th century. In any case, a musician's timing helps. In order to keep a smooth, even movement, I needed to grasp my UTA in both arms, in a hug, and move the telescope with my body, adjusting my head to keep my eye positioned — this became physically tiring as the night progressed.

From the book I also learned Herschel could move his telescope a small amount in azimuth and could track an object for about 15 minutes. This gave him more time to perceive and describe interesting objects. Also, Caroline did not rely on just a sidereal clock but primarily on Flamsteed's catalogue (volume 3 of the *Historia Coelestis Britannica*). Caroline would list out the stars which would be in the path of that night's sweep, and forewarn William of their coming; William would then describe the nebulae's distance and position from these reference stars. I lacked this second set of eyes to help with navigation. As the night progressed I realized that my elapsed time would not be enough to help locate an object later; since many of the objects appeared on the edges of the FOV, my elapsed time would be many seconds off. So I tried to remark whether I was near the top or bottom of the band, and to remark on any double star, red star (which might be a carbon star), or pattern of stars which might give me more reference points when reviewing the charts later. This has been of some help so far, but my progress is slow.

In all I recorded 175 observations. The majority of these were not bright, easily seen galaxies, but faint objects which I decided to describe in case later I could confirm them as galaxies. There were a large number of NGC, IC, and MGC objects in the band of the sweep, so it's probable I observed objects not on the Herschel Sprint list. I wanted to cover my bases by describing everything I had the slightest suspicion of being a galaxy. I separate the objects I observed into three categories:

High confidence: 56 objects were palpable hits, clearly visible as galaxies. Whether bright or faint, I was able to verbalize a complete description of the nucleus, core, halo, size, and position angle. I am in the process of identifying these on my charts.

Medium confidence: 62 objects I suspect are galaxies but I am not sure. My descriptions are typically "low surface brightness," "dim patch," "some glow with AV," "small, faint." My uncertainty comes from my lack of experience observing anything dimmer than a typical H400 list 11-12th apparent magnitude galaxy. These might be the usual fare for those used to observing faint objects; I am unfamiliar how they would appear in the eyepiece.

Low confidence: 57 objects are very uncertain. My descriptions are along the lines of "dim hazy star" or "mistiness." Too uncertain to have more than a slight suspicion they are galaxies. I'd need to find them on a chart to prove it to myself — and to gain more experience observing very faint objects to recognize them correctly.

It's frustrating not to have more certainty in what I saw, and I will spend the next few weeks settling a final tally. On the one hand, I had the desire to bear witness and "prove" I saw the objects by identifying them. But on the other hand, through the night I permitted myself to have that feeling of discovery. I can hear the excitement in my voice during playback when an obvious galaxy comes into view, such as at elapsed time 1:09:30: "real pretty one, right near the top of the band, glowing core, stellar nucleus with AV, elongated NE-SW, pretty one. Large. Spanning 1/3 FOV, fat, 3:2 a nice fat one" [I found this later to be **NGC 3486**]. Or at 2:45:39: "big, beautiful long one, bright core, you can see a dark lane, it's so long, oh my god and it's nearly out of FOV, I just barely got it. That's a nice one. 30 seconds too late" [this I found later was **NGC 4565**].

And that sums up the experience of visual astronomy for me. I feel the melancholy thrill of beauty fleetingly held; the intellectual astonishment trying to understand our true relation to the universe; the challenge of describing it to others so they may share. That Herschel was able to do these things so successfully speaks to his greatness. As I remarked to another observer, when doing the Herschel Sprint, you know you are in the presence of the Master.

SJAA Member Observation Posts

Tuttle Creek for Apr-2015 New Moon
From Gary Chock

I made a stargazing trek to Tuttle Creek this early spring for the Apr-2015 New Moon. Weather forecasts were good enough, but some dynamic conditions were possible. My schedule was flexible, so I played it by ear, adding extra nights mid-trip to make up for windy/cloudy nights.

For these multi-night observing trips, I am satisfied with one very good to excellent night and with a couple of other good to very good nights. You take in stride the difficult evenings with viewing through "sucker" holes between clouds or windy/cloudy nights that are good for catching up with sleep. I like the camping anyway. The Observing Report follows.

Tuttle Creek Campground is in the Alabama Hills near Lone Pine, CA. It is a dark rural sky site (Bortle Class 3, maybe 2) with large mountain ranges to the east and west, low horizons to the north and south. Airglow and zodiacal light are sometimes seen. There is a low light dome from Lone Pine to the east, but its effects are below the height of the Inyo Mountains and shielded by the Alabama Hills.

From San Jose, it is about 425 miles one-way, 8 to 9 hour drive to the campground (with rest stops). I took the route through Bakersfield, then south of the Sierra Nevada range, and North on 395 to Lone Pine, the nearest city for supplies. Lone Pine is only 15 minutes away from the campground - coffee shops and pizza to-go!

It is a nicely developed campground with reserved sites ($5 per night) and new-ish vault pit toilets. Potable (treated creek) water available at designated spigots (Mar through Oct), but I decided to bring my own water.

I spent the daytime lazily enjoying the campsite. Campsites are placed alongside Tuttle Creek, so I spent some time reading books with my feet dangling in the (cold) water. Some folks went to Lone Pine to hang out. There's also Alabama Hills, Cerro Gordo Mines, and Manzanar nearby to explore. In addition, the most impressive peaks of the Sierra Nevada are in view - Mt. Whitney, Lone Pine Peak, and Mt. Williamson.

It can be nice in the fall and spring months when Grandview Campground is inaccessible or too cold and Panamint Valley is too hot. It is 5,120 feet above sea level versus 8,600 feet for Grandview and 1,200 feet for Panamint Valley. During the summer, it can be hot and crowded. Note that there is a greater chance that nearby campers may have annoying lights.

<u>April 2015 New Moon</u>
I observed/camped for 6 nights (Wed 15-Apr-2015 to Tue 21-Apr-2015) at Tuttle Creek Campground along with 4 others.

I had a mixed bag stargazing-wise, but the whole experience with camping was nice. Wednesday night was very windy and gusty. The other nights had some Sierra afternoon thunder clouds in the early evening, then clearing. Thursday was good. Friday was very good. Saturday was okay. Sunday night was good. Then the last night on Monday had high overcast. Average SQM meter readings ranged from 21.3 to 21.7.

<u>Observing Report</u>
The best night was Friday with very good transparency and seeing. I concentrated on galaxy hunting in Ursa Major, Canes Venatici, Coma Berenicies, Virgo, Leo, Corvus, Hydra, and Centarus - studying between 40 to 50 objects well into the early morning. With my 10" Dob, I could make out objects just past magnitude 12. Multiple objects were wonderful in the giant binoculars - M81/M82, Leo Triplet, and Markarian Chain.

Highlights included NGC 4490 Cocoon Galaxy, NGC 4449 an irregular galaxy, NGC 4631 Whale Galaxy, NGC 4565 Needle Galaxy. The Markarian Chain was wonderful. I found many of the dimmer members and perused objects around it and Virgo A.

By the time Omega Centauri was high enough, it had rotated westward and was blocked by the southern Sierra Nevada range. Instead, I enjoyed studying NGC 5128 Centarus A, making out its "S" shaped dust lane. When M13 Hercules Globular Cluster was near zenith, it was almost as good as that one excellent night's view I enjoyed at Grandview last year.

Here's my stream of observations: M81/M82, M108, M97 Owl Nebula, M109, M101, M51, M63, M94, NGC 4490 Cocoon Galaxy, NGC 4449, M106, NGC 4631 Whale Galaxy, NGC 4889, NGC 4559, NGC 4565 Needle Galaxy, M64, M53, Leo Triplet (M65, M66, NGC 3628), M95, M96, M105, Virgo A, Markarian Chain ("Face" M84, M86, NGC 4387, NGC 4388, NGC 4402; "Eyes" NGC 4435, NGC 4438; NGC 4443, NGC 4458, NGC 4473, NGC 4477), M88, M91, M90, M89, M58, M104 Sombrero Galaxy, M68, M83, NGC 5128 Centarus A, M92, M13 Hercules Globular Cluster.

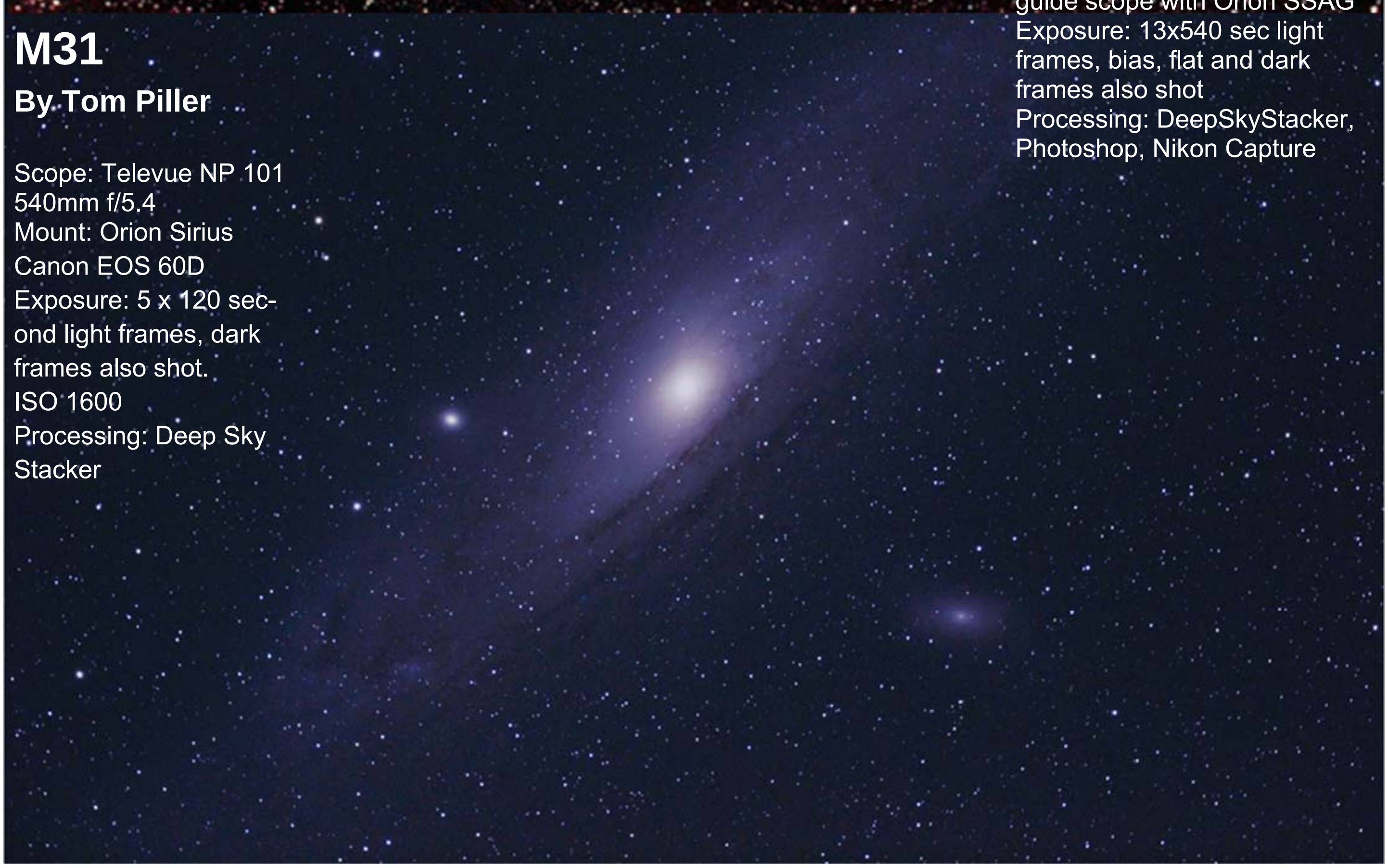
On a couple of early mornings, the Milky Way around Sagittarius and Scorpius was detailed with well-defined star clouds and dust lanes when viewed naked-eye and though binoculars. Also on one of those early mornings, I spent time observing Saturn when it was at its best for the trip. I could make out the Cassini Division and the gradation of the A-ring. Nice coloring in the polar region.

SJAA Astrophoto Gallery

California Nebula NGC1499

By Ed Wong

Scope: AstroTech AT72 with x0.8 reducer = 344mm F4.8
Camera: Canon T3i Full Spectrum ISO800
Mount: Orion Sirius
Auto Guider: Orion 50mm guide scope with Orion SSAG
Exposure: 13x540 sec light frames, bias, flat and dark frames also shot
Processing: DeepSkyStacker, Photoshop, Nikon Capture



M31

By Tom Piller

Scope: Televue NP 101 540mm f/5.4
Mount: Orion Sirius
Canon EOS 60D
Exposure: 5 x 120 second light frames, dark frames also shot.
ISO 1600
Processing: Deep Sky Stacker

SJAA Astrophoto Gallery

Moon Eclipse Photography Tips

From Jack Zeiders

Observers considering watching all or most of the upcoming eclipse of the moon may wish to consider taking a few photos as well. It is fairly easy and no special equipment is required though a tripod helps.

No matter what camera you may have, you will likely be able to take very nice pictures of the lunar eclipse. You don't need a special camera. Almost anything from an IPhone. Android, etc. to a DSLR, to film cameras work fine. You could even use a pinhole camera if you were so motivated. You don't need to track with shorter lenses, <50mm, though that would be handy if you wish to use a longer focal length lens or a telescope. I've had good results with a 200mm lens on just a tripod. A cable or remote release is helpful for the longer exposures during totality.

Remember that low ISO settings give less noisy results. ISO 100 is a good starting point. The exposure will change dramatically when going from fully illuminated to fully eclipsed. Exposure will vary with different cameras and lenses. You might wish to start at 1/200th @f /8 for the bright portions and add or subtract until it is right. Totality may range from a second or two to many seconds depending on how dark this eclipse is. Again, this is only offered as a starting point.

Note your where your lens focuses while the moon is bright as you can't focus on the eclipsed moon very well. Once you get good focus, turn off autofocus and put a piece of tape on the focus ring to keep it from getting bumped. For a beginner, I would suggest picking one focal length and sticking with that. If you are using a zoom lens make some test exposures using the focal lengths you plan to use during the eclipse. Many zooms don't focus at the infinity mark at all focal lengths. You may wish to test your lens.

You can make a series of exposures if you have bulb setting on one frame by simply mounting your camera on a tripod and covering and uncovering the lens at intervals. This sounds easy but is more difficult to do well than you may think.

For DSLR users, have a fully charged battery and plenty of room on your memory card. You won't have much time in totality, only about 5 minutes.

You also don't need to shoot a whole bunch of pictures unless you really want to. A half dozen shots will capture the various stages nicely leaving plenty of time to just watch the event.

Have fun !

Full moon rising near Lick Observatory
From Marilyn Perry

Location: Sierra Vista Parking lot on Sierra Road, San Jose
Date: May 3, 2015
Camera: Canon T3i
Camera Data: f/5.6, 1/15 sec, ISO 200, zoom lens at 67 mm
Post processing: Photoshop

Venus, Mars & Moon
From Vini Carter

Camera: cell phone

SOLAR TRANSITIONS OF THE STELLAR KIND

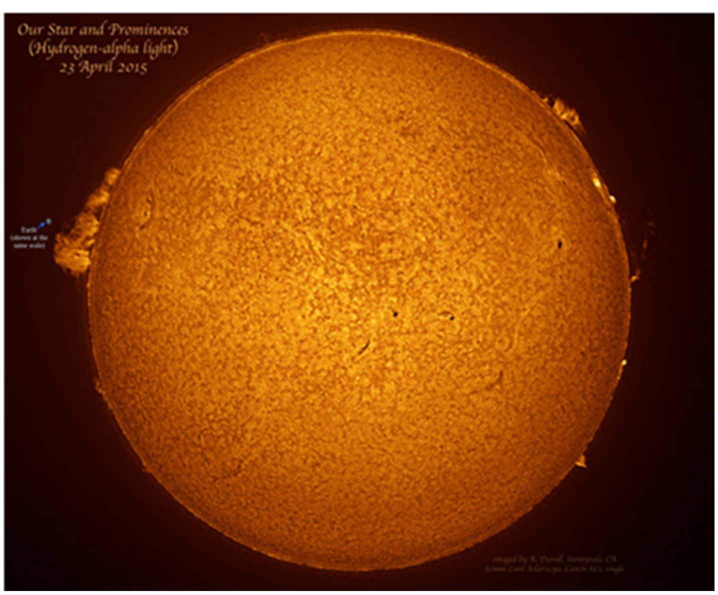
From Bill O'Neil

Editor's Note: The Month of May 2015 marked the start of Bill O'Neil's tenure over the SJAA SOLAR Program as Michael Packer steps down. Congratulations Michael on a stellar performance as program director, your efforts were and are greatly appreciated! Piller

Michael Packer did all the research on Solar Telescopes and the club acquired their Lunt 100mm Hydrogen-alpha telescope early in 2012. (H-a is used to look at solar flares and prominences on the edge of the Sun's disk). The first light Solar Sunday was Feb 5, 2012 (and was the same day as a Super Bowl). Michael created and ran a very successful SJAA Solar Program for over 3 years. He was recently elected as SJAA Vice President, so he is turning the Solar Program over to Bill O'Neil.

Bill is a Director of SJAA and has been active in Solar Sundays for years. He and his wife, Susan, were Astronomy Park Rangers (volunteers) up at Glacier National Park in Montana in the summer of 2013. In addition to a night time Star Party, they ran a daily Solar Program at Glacier using the NASA "Our Magnetic Sun" resources with park visitors, young and old. Bill looks forward to creating new Solar Education and Outreach opportunities to raise awareness of SJAA, and our excellent Solar Astronomy program.

Michael would like to thank Bill and SJAA members: Terry Kahl, Wolf Witt, Carl Reisinger, Kevin Lahey, Isaac Kikawada, Robert Duvall, Paul Mancuso, and Teruo Utsumi for their support and bringing out solar telescopes to public events.

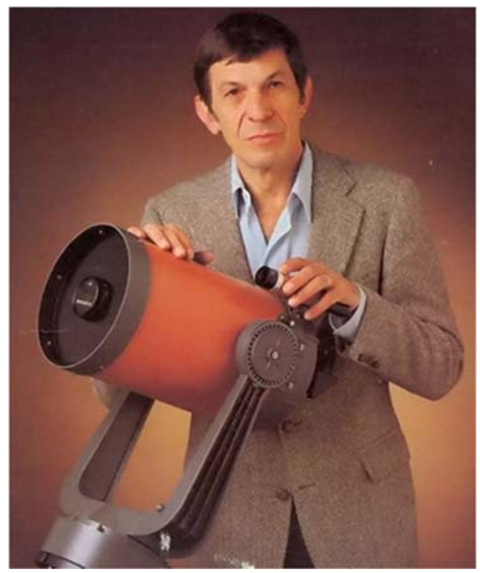


Recent Photo of SOL. Photo Credit Robert Duvall

Solar Sunday March 1st In Honor of Leonard Nimoy

Sunspot count 54 (SDIC) C-Class Flare Observed Live in H-Alpha

From Michael Packer



Lenard Nimoy 1931—2015

We talked about Leonard Nimoy at today's Solar Sunday. There is no doubt that Star Trek is part of the amateur astronomy mind set. The common element of wanting to seek out and explore our universe, and of course imaging life out there is intriguing. And we amateurs get to explore and imagine with our telescopes. Just as appealing – is imagining Gene Roddenberry's future – where there is no discrimination between a community of planets. Spock's character brought that universe to life and Leonard Nemoy's sensibility to the character endeared the minds of a staggering and still growing fan base that started 50 years ago. Many thanks Leonard for ensuring these ideals live long and prosper.

Sunday Solar had about 15 visitors today with many of them observing and talking about the sun for an hour or more. We counted 53 sunspots, 11 filaments (prominences), and limb prominences at moderately high power. Above are some of the dedicated patrons!

SJAA MEMBERS ATTEND GOOGLE STAR PARTY

Pictured: *Front rows of telescopes at the Google Star Party*
Photo Credit: Photo provided by Bill O'Neil

On Tuesday, Feb. 10, 2015, Google hosted a huge Star Party for its employees, family and friends. Over 800 people attended the event at Garfield Park (across from the Google Campus in Mountain View). There were over 40 telescopes and astronomers from many Bay Area astronomy clubs, including SJAA, Eastbay Astronomy, Tri-Valley Astronomers, Peninsula Astronomical Society, San Mateo County Astronomical Society, San Francisco Amateur Astronomers and Mount Diablo Astronomical Society.

Thanks to SJAA members: Prabath Gunawardane (and Google employee), Teruo Utsumi, Wolf Witt, Paul Colby and Marion Barker, Les Murayama, Hitesh Dholakia, Bill O'Neil and Rob Jaworski for signing up to support this event.

In addition to telescopes, there were booths and information tables set up by Space Science Labs at UC Berkeley (they brought a Planetarium and a 'Jeopardy' type game with push buttons and lots of Astronomy questions for kids), and the Chabot Space & Science Center had several Hands-On activities for young people.

Dr. Sandra Faber (Astronomer and 2013 recipient of the National Medal of Science from President Obama (and speaker at SJAA ;-) and Bob Kibrick (Director of Friends of Lick, who worked as Research Astronomer at Lick for over 30 years) were there to announce Google's $1 Million donation to Lick Observatory. (SJAA is having a Lick Tour and Fundraiser on May 27) In 2013 UC announced it would reduce funding for Lick starting in 2016, with a complete cutoff after 2018. While later reversing that decision, the observatory's budget remains is tight and relies on much of its financial support from UC and other donors. Google stepped up!

The Star Party received Rave reviews with comments like: "Best Party at Google Ever!" (Could it have been the free Pizza?); "The event was a smashing success"; What was your favorite part? "Seeing a real planet and three moons in a real telescope" (from 5 year old boy); "My kid didn't want to leave at the end!"

Pictured: Back rows of telescopes with Google Campus in background.
Photo Credit: Photo provided by Bill O'Neil

Annual SJAA Auction Sunday, March 8, 2015

From Michael Packer

35th Annual Spring Auction

It was "good to see friends from other clubs like Tri-Valley, Mid-Peninsula, etc." – SJAA's Bill O'Neil

First A Little History

Computer Auction Organizer JVN over a decade ago. Credit Dan Wright.

Kevin Medlock started off the auction with a little history. The San Jose Astronomical Association's First Auction was held 35 years ago at a Red Cross building in Los Gatos. The Red Cross is no longer there but the building still is. And there was a large crowd anticipating this first auction. But back then there was no computer or pizza to make the day go easier. In later years when SJAA did finally have an IBM computer to keep track of buys and sells for 100's of items it was still a slow process. Jim Van Nuland recalls one seller who brought in over a 100 items(!) and then accidentally put one on the auction block at a low price. He had to buy it back by out-bidding everyone else.

Upfront Thanks

35 years later, SJAA has better computers but generations of new volunteer Board members and "better" software still can make the day a lot of work for our organizers. A special thanks to the SJAA Board, our Auctioneer Kevin Medlock and snack organizer Marianne Damon. You guys made it real, fun, made it real fun!

Results

This year we had 156 items for auction. SJAA sold $3,749 worth of stuff and Sellers sold $4,887. Congratulations to both Buyers and Sellers. Below are a few pics of the day.

Thanks to all of you for joining – see you in the field and at next auction!

Photo Credit for the pictures below and on following pages goes to Ed Wong; there are some great shots.

Annual SJAA Auction Sunday, March 8, 2015

Annual SJAA Auction Sunday, March 8, 2015

KID SPOT

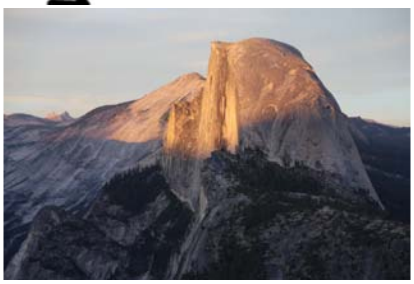


Image: Half Dome
Photo Credit: SJAA

Kid Spot Jokes:

- **Why does the moon go to the bank?**
  To change its quarters

- **What is the center of gravity?**
  The letter v.

Kid Spot Quiz:

A. Which space shuttle carried Hubble to orbit in April 1990?
B. How high is Hubble over the Earth's orbit?

Did You Know?

Source: Astronomy Magazine March 2015 issue
"500 Coolest Things"

◊ Exactly 88 constellations cover the sky with no gaps and no overlaps. *(M. Packer also noted: 88 is also the same number of keys on the modern piano...Music of the spheres?)*

◊ The solar system's largest moon, Ganymede, has 25% more volume than Mercury.

◊ At a distance of 4.2 light years, Proxima Centauri (Alpha Centauri C) is the closest star to the Sun.

◊ Of the 9,113 official features on the Moon, a mere 421 (4.6 percent) are not craters.

◊ Two globular clusters vie for the title of nearest to Earth: M4 in Scorpius and NGC 6397 in Ara. Each lies 7,200 light-years away.

◊ In 2014, scientists studying the jovian moon Europa discovered evidence of plate tectonics, a first for a world beyond Earth.

◊ Saturn's moon Enceladus likely has a large subsurface sea that feeds the geysers shooting from its south polar region.

Kid Spot Quiz Answers:

1. Discovery

2. 347 miles

SJAA Library!

SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!
Please check out our wish list on the SJAA webpage:

http://www.sjaa.net/sjaa-library/

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

Big thanks to go Ed Wong, Phil Chambers, and Dave Ittner for being the gear experts who faithfully make themselves available at Fix It day along with Vini Carter who has stepped up to take over as Fix It program chair from Ed Wong.

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are held the 1st Sunday of every Month from 2:00-4:00 PM at Houge Park weather permitting. See article and photos on page 7.

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions will now be held every other month. Next one is scheduled for June. In addition to the indoor sessions, Ken Presti has offered to hold an outdoor session for the QS attendees to help them learn how to use the loaner scopes and the night sky.

The first field session was held on Saturday, April 18th. There were 13 scopes out at RCDO. Everyone enjoyed the session.

Since it makes no sense to have a Quick Start scope sitting in the storage closet at Houge Park, it is my intention to offer these scopes out to our members with preference to past QS attendees.

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects. We'll need extra help to pull this off, but it appears to be doable."

The announcement and background can be found here:
https://groups.google.com/forum/#!topic/sjaa-announcements/ScBqsdpNmzg

Loaner Program

The purpose of this program, headed up by Manoj Koushik, is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment below. Please note that certain items have restrictions or special conditions that must be met.
If you are an SJAA member and an experienced observer or have been through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.

SJAA Contacts

President:	Dave Ittner
Vice President:	Michael Packer
Treasurer:	Rob Jaworski
Secretary:	Teruo Utsumi
Director:	Greg Claytor
Director:	Lee Hoglan
Director:	Ed Wong
Director:	Bill O'Neil
Director:	open
Memberships:	Ed Wong
Beginner Class:	pending
Fix-it Program:	Vini Carter
Imaging SIG:	Mark Striebeck
Library:	Sukhada Palav
Loaner Program:	Manoj Koushik
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Publicity:	Rob Jaworski
Questions:	Lee Hoglan
Quick STARt	Dave Ittner
Solar:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Teruo Utsumi
E-mails:	

http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
http://www.sjaa.net/contact

From the Board of Directors

Announcements

The newly elected Board Officers:

President:	Dave Ittner
Vice President:	Michael Packer
Treasurer:	Rob Jaworski
Secretary:	Teruo Utsumi

Bill O'Neil takes over the club Solar Program as Michael Packer steps down.

The Club's annual Pinnacles observing event will take place Friday and Saturday nights July 10th and 11th. By Reservation only.

2015 GSSP will take place July 15th thru 19th.

2015 Cal Star will be held Sept 10th thru 13th.

Bino star gazing Saturday, 7/25, 8/22

One Board seat remains open.

Board Meeting Excerpts

March 07 2015

In attendance

In attendance: Rob Jaworski, Lee Hoglan, Michael Packer, Dave Ittner, Ed Wong, Teruo Utsumi
Absent (excused): Bill O'Neil (proxy to Dave), Greg Claytor
Guests: None

Battleship Binocs, Other Donated Items

Per Dave Ittner We have had possession of the naval binocular for some years now without any progress of its divestment. Dave reported that he received a bid of less than $2,000. Some on the board felt the bid is undervalued. Lee agreed to take the lead in divestment of the naval binocular. The board agreed to give Lee three months, after which the board will revisit this issue at the June 2015 board meeting.

Dave presented a general procedure to divest items acquired by the club. The board authorized Dave to divest such acquired equipment for items valued under $1000 and to get board approval for items at or above $1000.

Goals for 2015

Dave discussed his goals for the board in the coming year. There is little documentation for most of the roles and tasks involved in running the club and the various events. The main goal is to put procedures and their documentation in place for future use.

Forum and Meetup Group

Dave stated that there has been much discussion the last several weeks about Meetup for communication of club events to members and the general public.
Rob makes a motion to start an SJAA specific Meetup group and a budget of $200. Dave seconds, approved by all.

Messier Half Marathon

Teruo is organizing and Ed is Mendoza permit holder. Scheduled date is TBD.

April 04 2015

In attendance

Rob Jaworski, Lee Hoglan, Dave Ittner, Ed Wong, Bill O'Neil, Teruo Utsumi
Absent: excused – Michael Packer; unexcused – Greg Claytor
Visitor: Manoj Koushik, Vini Carter, Leslie Carter, Marianne Damon

Presidents Message

Significant achievements
Dave talked about his goals/priorities.

- Complete CoSJ proposal – A very high priority for the club is completion of the proposal which is due next year as required by the City of San Jose. One major effort that will be required is documenting how many are served by SJAA functions. There has been ongoing discussion as to how to implement this.
- Document roles and tasks for club functions – Ease transition for new people coming into roles and tasks.

Improve communications from club to members and general public
Need to revamp club website.
We've recently started using Meetup.
Leslie contacted the local community center about promoting SJAA activities. She is looking into different avenues of marketing the club, e.g., postings, submission of event notices to their newsletter.

High Speed Internet at Houge Park

Sukhada contacted ATT and Comcast about getting Internet to Houge Park Bldg. 1. They are scheduled to make surveys this week. Sukhada to report on results of surveys and future steps.

Board/Member Involvement

Teruo talked about the various tasks involved with running the General Meetings and requested help.

Lick observatory visit

Bill reported the Lick trip is scheduled for Wed. May 27 (was originally 5/20). The cost is $50 per person ($20 fee + $30 donation). We expect to raise $1,500 total for Lick. The financial obligation of the club for the trip is the minimum of $550 required by Lick which we expect to be paid by participants. Bill to continue to work out details.
Bill makes a motion to approve the trip and to front the minimum $550 cost. Lee seconds, approved by all.

Pinnacles Trip

Ed reserved the group campsites at the Pinnacles east campground for July 10, 11. The maximum number is 40 people. Ed to work out details.

May 02 2015

In attendance

Michael Packer, Rob Jaworski, Lee Hoglan, Ed Wong, Bill O'Neil, Greg Claytor, Teruo Utsumi
Absent: excused - Dave Ittner (proxy to Michael)
Visitor: Vini Carter, Leslie Carter, Sukhada, Manoj Koushik, Tracy, Marianne Damon

SJAA Hotline

The board agreed to terminate the SJAA phone hotline. Follow up by Michael, Rob.

Internet at Houge Park

Sukhada looked into high-speed internet at Houge Park and arranged for ATT to install a DSL line. Informally the Youth Shakespeare Group has agreed to split the monthly cost.

Advertising Budget

Rob noted the $200/quarter publicity budget is not sufficient to do what he'd like. His idea is to use a big SJAA horizontal banner, smaller vertical banner in order to target the local neighborhood. Rob makes motion for $2000 for this year, for local publicity. The motion was approved.

Advanced Loaner Program - Manoj

Manoj gave his observation that we need more beginner-friendly equipment. He presented a proposal to purchase two large aperture binoculars and mounts.
Manoj makes motion for $1100 to purchase two sets of binoculars and mounts which was approved by all.

Place
postage
here

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type:

☐ **New** ☐ **Renewal** (Name only if no corrections)

☐ **$20** Regular Membership with **online** Ephemeris

☐ **$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: **http://www.sjaa.net/sjaa-newsletter-ephemeris/**

Questions? Send e-mail to: **sjaamemberships@gmail.com**

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: **http://www.sjaa.net/join-the-sjaa/**

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